

Comparing the Performance of Leading VQE Algorithms for Computing Ground-State Energies of Amino Acids

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Abstract

Simulating molecules is a major application of quantum computing, with the potential to overcome exponential scaling constraints of classical computation. Researchers use different methods (various ansatz, active space truncation methods, etc.) in order to evaluate the readiness of NISQ computers in order to test current simulation capabilities. We present an integrated repository of over 10 different ansatz from published papers and two different truncation methods (customizable to adapt molecules to larger qubit counts), applicable to any set of mapped hamiltonians, providing a single pipeline for comparing performance along different axes (variance, computational time, etc.). We apply them to simulate different amino acids, using hamiltonians taken from the QMProt Dataset. We have then run multiple experiments, such as (i) quantifying noise resilience: optimizing the same hardware-efficient ansatz under identical initialization while sweeping PennyLane noise channels and strengths, and measuring parameter drift, cosine similarity of optimal parameters, and energies evaluated on noiseless versus noisy backends; (ii) studying barren-plateau-related trainability via gradient-variance diagnostics and optimization trajectories across initialization strategies and ansatz depth on small systems; (iii) comparing adaptive versus fixed ansätze at matched parameter budgets, reporting outer-loop iterations, wall time, and especially total cost-function evaluations to fairly contrast greedy adaptive growth with layered hardware-efficient circuits; and (iv) mapping accuracy versus expressive capacity by sweeping the number of retained adaptive operators (equivalently parameters) and recording ground-state energy error relative to classical references.

1 Introduction

The accurate simulation of molecular systems remains one of the most formidable challenges in computational science. At the heart of this challenge lies the Electronic Structure Problem: solving the Schrödinger equation to determine the ground-state energy (GSE) of a many-electron system. Classical methods, while sophisticated, face an exponential scaling wall; as the molecular system size increases, the number of possible electronic configurations grows combinatorially, rendering exact solutions for large biological molecules computationally intractable. Quantum methods, on the other hand, represent wavefunctions in a Hilbert space that scales linearly with the number of qubits.

In the current Noisy Intermediate-Scale Quantum (NISQ) era, traditional algorithms like Quantum Phase Estimation (QPE) remain impractical due to their requirement for high-depth circuits and error correction [Preskill, 2018]. To circumvent these hardware limitations, the Variational Quantum Eigensolver (VQE) was introduced as a hybrid quantum-classical algorithm [Peruzzo et al., 2014]. VQE leverages the variational principle, which states that the expectation value of the Hamiltonian H for any trial wavefunction is an upper bound to the true ground-state energy E_0 :

By offloading the parameter optimization to a classical optimizer and using the quantum processor solely for state preparation and measurement, VQE significantly reduces the coherence time requirements [McClean

et al., 2016]. Since its inception, VQE has been successfully applied to small molecules such as H_2 , LiH , and BeH_2 [Kandala et al., 2017]. The transition from small "toy" molecules to biologically relevant systems represents the next frontier for quantum advantage. In drug discovery, the efficacy of a lead compound is often determined by its binding affinity, which is deeply rooted in the GSE of the protein-ligand complex [Mullane Srimani, 2020].

Understanding the energy landscapes of amino acids and their transition states is vital for modeling enzymatic catalysis and protein folding. Precise calculations of molecular interactions allow for the characterization of non-covalent bonding and electrostatic potential surfaces that classical force fields often oversimplify [Outeiro et al., 2023]. However, the biological scale introduces high-dimensional Hamiltonians, necessitating VQE algorithms that are not only accurate but also computationally efficient in terms of measurement overhead and circuit depth.

Classical methods suffer from the exponential scaling of computation costs; in particular, strongly correlated systems "pose a significant challenge for classical quantum chemistry approaches, necessitating multiconfigurational electronic wavefunctions" [Vaquero-Sabeter et al., 2024]. Furthermore, recent developments in VQEs suggest promising results in protein folding simulation, potentially opening opportunities for drug discovery against disordered proteins long considered difficult to target. [Uttarkar et al., 2026] The rapid proliferation of VQE variants has created a "zoo" of algorithms, including: Chemically-inspired Ansatz: Such as Unitary Coupled Cluster (UCCSD), which provide high accuracy but often result in prohibitively deep circuits [Anand et al., 2022]. Hardware-Efficient Ansatz (HEA): Which prioritize low circuit depth at the cost of being susceptible to "Barren Plateaus" during optimization [McClean et al., 2018]. Adaptive Ansatz: Such as ADAPT-VQE, which grows the operator pool dynamically to balance accuracy and efficiency [Grimsley et al., 2019]. Despite this innovation, the field lacks a unified metric for comparison. Most studies benchmark algorithms on different hardware, using different optimizers, or on varying molecular geometries. This fragmentation makes it difficult for researchers to select the optimal algorithm for specific biological applications, such as calculating transition-state energies where the error tolerance is extremely narrow [Abidi et al., 2024]. To address these challenges, this study utilizes the newly released QMProt database [1]. Unlike previous datasets focused on small organic molecules (e.g., QM9), QMProt provides the mapped Hamiltonians specifically for the core amino acids, providing a standardized "battleground" for quantum algorithms in a biological context. We present a comprehensive framework designed to evaluate VQE performance across three critical axes:

- Accuracy: Proximity to the exact diagonalization, or FCI, limit within chemical accuracy, defined as 1.6 mHa.
- Computational speed: Number of iterations and total shots required to reach convergence.
- Error spread: Robustness of the algorithm against statistical noise and shot-noise fluctuations.

By applying this framework to the amino acids within QMProt, we aim to provide a roadmap for the scalable simulation of larger biomolecules, ultimately bridging the gap between quantum theory and practical pharmaceutical application.

2 Methodology

We exclusively used the QMProt Dataset. They compute Hamiltonians using OpenFermion with inputs including coordinates, charge, spin (via multiplicity), and a basis set, then run PySCF with SCF, and finally output a fermionic Hamiltonian using Jordan-Wigner. They used the RHF method with a minimal basis set (STO-3G in most cases). They then extracted the 3D atomic coordinates from PubChem SDF files.

There are 45 molecules in the dataset, including water, the amino group, the main amino acids, and their radicals. A distribution of their reference ground state energies, as well as qubit mappings, and electron representations can be seen in Figure 1 and Figure 2 respectively.

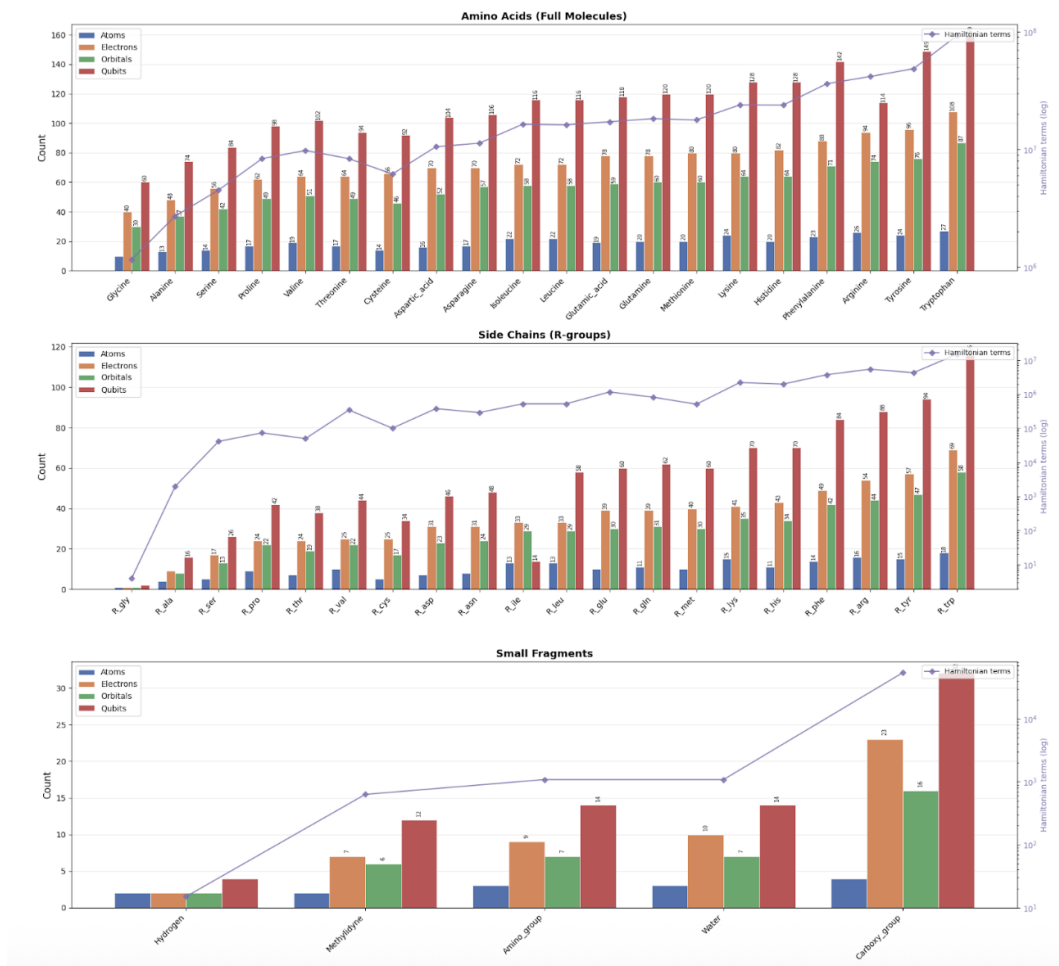


Figure 1: Number of atoms, electrons, orbitals, and qubits for the molecules available in the QMProt dataset.

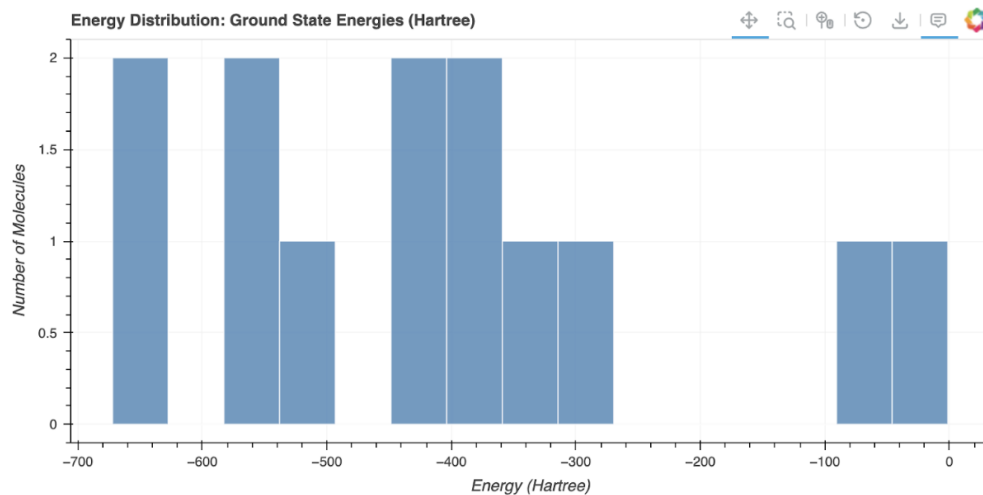


Figure 2: Ground state energies (reference) from the dataset.

There have been dozens of different VQE algorithms developed by researchers over the last few years. For this repository, we focused on major ones present in the academic and industry communities. A list of them can be found in Table 1. We used available code from the researchers’ GitHubs, and adapted them to run on the PennyLane/Qiskit ecosystem so they could all be run from our centralized repository.

This work benchmarks VQE variants under an explicit qubit budget with two possible truncation methods implemented. The goal is to preserve the highest energetic contributions while reducing circuit width and measurement cost, enabling consistent comparisons across algorithms on NISQ-relevant problem sizes. Many quantum chemistry Hamiltonians, once mapped to qubits, contain a large number of Pauli strings with small coefficients. For example, phenylalanine from the dataset is default mapped to 132 qubits. Using hard cutoffs on coefficient magnitude can help reduce to a substantially smaller effective Hamiltonian while preserving the main optimization landscape [2].

Near-term benchmarking is constrained by limited qubit counts and the depth of circuits that current hardware can reliably execute, to such an extent that directly representing large molecular systems is infeasible for the time being [3]. However, by reducing the number of qubits required to map a system, we’re reducing our problems to a complexity that we can represent on current hardware. A common strategy is to reduce the effective qubit space by focusing on the most chemically relevant orbitals. Another approach is to take advantage of symmetries in fermionic Hamiltonians to eliminate redundancies and lower qubit requirements without altering the problem [4]. Regardless, fidelity is also important, and through methods such as extrapolation to the zero noise limit, the effects of errors and decoherence can be mitigated [5]. Such reductions, together with validity-preserving schemes, enable VQEs to converge on near-chemical accuracy using circuits shallow enough for NISQ devices.

2.1 Active Space Truncation

To enable benchmarking under tight qubit constraints to allow simulation, we used a systematic active-space truncation protocol to reduce the electronic Hamiltonian prior to qubit mapping. All electronic structure calculations were performed using PySCF. For each molecule, we first computed Hartree-Fock molecular orbitals and orbital energies. Core orbitals were identified and frozen based on energetic separation and chemical considerations (standard from the library). Candidate active orbitals were proposed using a combination of orbital energy proximity to the Fermi level, natural orbital occupation diagnostics from correlated calculations, and automated selection strategies such as the Atomic Valence Active Space (AVAS) method implemented in PySCF [6]. This procedure defined an active space characterized by a specified number of active electrons and spatial orbitals.

The active space was validated using multi-configurational calculations within PySCF. Complete active-space configuration interaction (CASCI) or complete active-space self-consistent field (CASSCF) calculations were performed to assess whether the selected orbitals captured the dominant static correlation effects [7, 8]. Validation criteria included the presence of significant configuration mixing and fractional natural occupation numbers within the active subspace.

After validation, a reduced second-quantized Hamiltonian was constructed in the active orbital basis. Frozen-core and external orbitals were excluded, yielding a fermionic Hamiltonian restricted to the active space. This reduced Hamiltonian was generated using either Qiskit Nature’s active-space transformation utilities or the OpenFermion–PySCF interface. The resulting fermionic operator was then mapped to qubits via standard fermion-to-qubit transformations. The final qubit count is twice the number of spatial active orbitals, making sure that qubit reduction occurred at the physically meaningful fermionic level before any of the VQEs.

The mathematical background for this is as follows:

- The pipeline starts with the full Hamiltonian input.

Table 1: List of VQEs not adapted into the repository due to missing code.

VQE	Paper and Code	Description
CB-VQE (Penny-Lane)	Paper: None written Code: https://www.pennylane.ai/qml/demos/tutorial_classically_boosted_vqe/	Enhances standard VQEs by performing a classical post-processing step that projects the molecular Hamiltonian onto a two-dimensional subspace spanned by the Hartree-Fock reference state and the VQE-optimized state. It then solves the generalized eigenvalue problem to guarantee a lower or equal energy compared to VQE, while requiring only Hadamard test circuits to compute cross-terms. This is valuable for near-term quantum devices where circuit depth and noise are limited.
NN-VQA	Paper: https://doi.org/10.1103/PhysRevApplied.21.014053 Code: https://github.com/Miao-JQ/NN-VQA	Uses neural-network-encoded variational quantum algorithms to address the challenges of implementing VQAs on NISQ devices.
CS-VQE	Paper: https://quantum-journal.org/papers/q-2021-05-14-456/ Code: https://github.com/wmkirby1/ContextualSubspaceVQE	A hybrid algorithm that reduces the number of qubits needed to achieve chemical accuracy. It works by mathematically dividing a molecule’s energy calculation into a classical portion that is easily solved on a regular computer and a strictly quantum portion, called the contextual subspace. VQE is then run only on the contextual subspace.
iQCC	Paper: https://pubs.acs.org/doi/10.1021/acs.jctc.9b01084 Code: https://github.com/sandbox-quantum/Tangelo-Examples/blob/main/examples/variational_methods/iqcc_using_clifford.ipynb	Uses constant-depth, shallow quantum circuits that are iteratively updated through canonical transformations of the Hamiltonian on a classical computer. iQCC significantly reduces circuit depth, making it less prone to errors on NISQ devices.
VAns	Paper: https://arxiv.org/abs/2103.06712 Code: https://github.com/matibilkis/qvans	A variational ansatz architecture search method that adaptively builds quantum circuits by adding, removing, and optimizing gates. It aims to find compact, problem-suitable ansatz circuits rather than relying on a fixed ansatz structure.
Hamiltonian Variational Ansatz	Paper: None written Code: https://quantum.cloud.ibm.com/docs/en/api/qiskit/qiskit.circuit.library.hamiltonian_variational_ansatz	A physics-informed ansatz that turns the structure of a molecular Hamiltonian into a quantum circuit by partitioning Hamiltonian terms into commuting groups and implementing time-evolution operators for each group. It mirrors the problem Hamiltonian, leading to fewer variational parameters while maintaining strong expressibility for ground-state preparation. This can reduce landscape complexity, improve trainability, and guide ansatz design.

Table 2: Summary of additional VQE methods adapted into the repository.

VQE	Paper and Code	What does the VQE do?
VQE-UCC	Paper: https://doi.org/10.3389/fphy.2021.735321 Code: None available	Implements the Unitary Coupled Cluster Singles and Doubles ansatz with the SP-UCC spin-symmetry optimization from Guo and Chen (2021). This pairs alpha/beta single excitations under a shared amplitude and restricts double excitations to pair excitations, where both spins of one spatial orbital are excited together. This reduces the parameter count from $O(M^4)$ to $O(M^2)$ while maintaining near-UCCSD accuracy for singlet ground states. The circuit also applies SingleExcitation and DoubleExcitation Givens rotations to the Hartree-Fock reference state, where all-zero parameters recover the Hartree-Fock energy exactly. In our benchmarks, it consistently matches ADAPT-VQE accuracy, for example 0.1 mHa on ARG, with only 8 parameters on 8 qubits, the fewest of any algorithm tested.
QAOA-Inspired	Paper: https://www.nature.com/articles/s41534-025-00982-6 Reference Code: https://pypi.org/project/bqs/	Uses a QAOA-style ansatz with alternating cost and mixer layers. A qubit Hamiltonian object is initialized with the number of QAOA layers, p , and a mixer type, such as X or XY. Each layer contains a cost parameter, γ , and a mixer parameter, β . For the cost layer, the circuit applies an evolution of the form $e^{-i\gamma H_{\text{cost}}}$, implemented through rotations based on the Hamiltonian structure. Noise is then applied after each layer, with no effect in the statevector setting.
NN-AE-VQE	Paper: https://arxiv.org/abs/2411.15667 Code: None available	Compresses the VQE ansatz through a quantum-autoencoder bottleneck. An encoder maps the full n -qubit Hartree-Fock state into $n/2$ latent qubits, a compact latent ansatz performs variational optimization in the compressed subspace, and a decoder reconstructs the full-space trial state. This reduces the number of variational parameters by roughly half compared to applying the same ansatz over all qubits. A classical neural network can additionally predict near-optimal circuit parameters to bypass expensive optimization loops. The method also supports configurable encoder, decoder, and latent styles with a built-in architecture search over style combinations.

- HF calculated for an orbital basis and a reference occupancy pattern.
- MP2 diagnoses which orbitals are no longer “cleanly” 0 or 2 occupied, using a perturbative method around a single-reference HF state.
- The one-particle reduced density matrix, 1-RDM, is calculated and diagonalized.
- Correlation score highest for around 1, or half occupied.
- Orbitals are partitioned. Frozen core, for doubly-occupied orbitals, assumes certain low-energy orbitals remain doubly occupied in every determinant considered.
- Active space electrons calculated.
- Validating that reduced model with CASCI/CASSCF.
- Mapping the resulting fermionic Hamiltonian to qubits with Jordan-Wigner, for consistency with the original paper.

In our pipeline, we’ve also implemented functions that allow researchers to visualize the chosen active spaces in an interactive manner. Figure 3 displays the active spaces chosen for a few model amino acids.

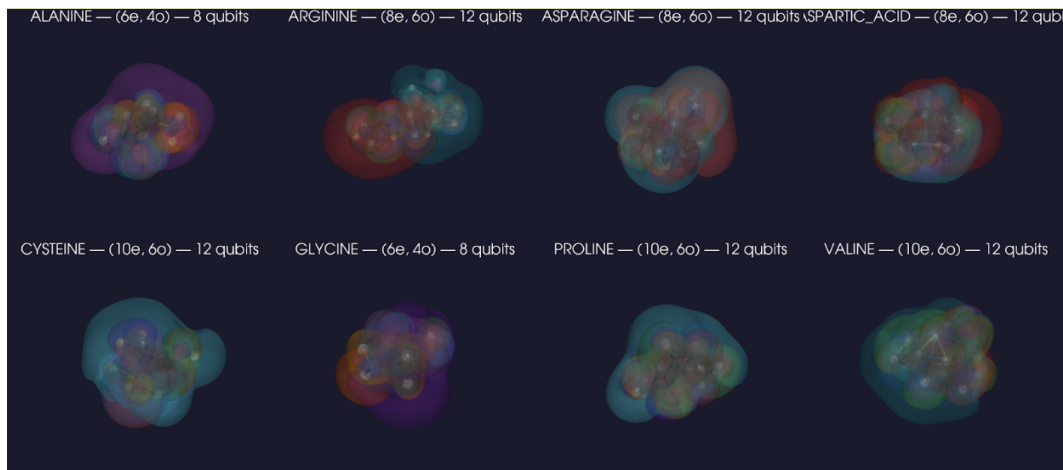


Figure 3: Active orbitals of model amino acids, visualized.

We also have a flag to enable constructing a reduced (truncated) qubit Hamiltonian that retains only a subset of Pauli terms from the full mapped Hamiltonian.

2.2 Contextual Subspace Reduction

To mitigate the high qubit requirements associated with amino acid Hamiltonians, we implemented an optional Contextual Subspace (CS) reduction step. Based on the theory proposed by Kirby et al., this method partitions the Hamiltonian (H) into a classical component and a quantum component. This was integrated into the repository using code inspired from the repository used in the paper [9].

The reduction process follows a three-step workflow:

1. Sub-Hamiltonian extraction: We identify the largest noncontextual sub-Hamiltonian, H_{NC} , which consists of a subset of Pauli terms whose expectation values can be simultaneously determined by a classical hidden-variable model. This extraction uses a greedy depth-first search, DFS, prioritized by the magnitude of the coefficients.

2. Classical optimization: The ground state of H_{NC} is solved classically. We use a hybrid optimization strategy with exhaustive enumeration of all 2^m sign assignments for the commuting sector and differential evolution for the anti-commuting sector.
3. Qubit projection: The classical solution provides m stabilizer constraints. We apply a sequence of Clifford rotations to map these constraints onto single-qubit Z operators. These stabilized qubits are then projected out, resulting in a reduced effective Hamiltonian acting on as few as $n - m$ qubits, for example, reducing a 10-qubit water Hamiltonian to 4 qubits. Intermediate qubit targets are supported by releasing generators back to the quantum simulation.

2.3 Basis Transformation and State Preparation

The Clifford rotations used in CS reduction shift the system into a noncontextual qubit basis. This necessitates a corresponding transformation of the Hartree-Fock (HF) initial state to ensure physical consistency.

Our framework utilizes a `_prepare_initial_state()` function that applies the identical unitary transformation used in the Hamiltonian reduction to the initial $|HF\rangle$ state. This transformed state is then projected onto the reduced qubit subspace. This ensures that the VQE optimization begins at a chemically relevant starting point, even within the transformed Hilbert space.

2.4 Framework Integration

A core feature of our framework is its ability to handle different VQE variants in the reduced subspace without manual reconfiguration. Algorithms are categorized into two primary logic flows:

- **Hardware-Native Algorithms:** Methods such as Hardware-Efficient VQE and QAOA-inspired VQE are treated as fully native. These depend only on the number of qubits and do not rely on an assumed fermionic structure, allowing them to operate directly on the reduced Hamiltonian.
- **Fermionic-Adapted Algorithms:** For algorithms that typically assume an orbital structure (e.g., Vanilla VQE, ADAPT-VQE, IQCC), the framework clears `n_electrons` and `n_orbitals` as a safeguard, since these quantities are undefined in the rotated basis.

In the case of Qubit-ADAPT, the standard "occupied-to-virtual" operator pool becomes physically meaningless after the Clifford basis rotation. To maintain expressivity, our framework automatically switches the operator pool to a full Pauli pool whenever CS reduction is active. This allows the adaptive algorithm to explore the transformed space effectively despite the loss of traditional molecular orbital structure.

One strategy we used to ensure that our VQE solutions for GSE were accurate is by making sure the reference state outputted the correct Hartree Fock energy, and setting all parameters to zero at the HF. We verified correctness of our variational implementations by computing the Hartree-Fock reference energy $\langle HF | H | HF \rangle$ and confirming that optimized VQE energies were not greater than this value (in noiseless simulations). Because the Hartree-Fock state is contained within the ansatz parameter space, the variational principle guarantees that the optimized energy is bounded above by the Hartree-Fock energy.

2.5 Experiments Run

- **Noise resilience:** Same molecule, same ansatz depth and shared initial parameters; optimized with `hardware_efficient_vqe` and PennyLane noise models while sweeping noise channel and strength p . Comparing noisy optimum vs noiseless baseline on energy and parameter drift (L_2 , cosine); optional batch runs across molecules with aggregated heatmaps.
- **Barren plateaus / initialization:** Hardware-efficient ansatz at several depths:
 - Part A: gradient variance or mean vs gradient vs depth over random initializations and named initial strategies

- Part B: repeated COBYLA runs comparing convergence and final energy across strategies at fixed depth.
- Trainability (matched parameters): For each target parameter count, e.g. 10 / 50 / 100, ran `qubit_adapt_vqe` fixed max operators vs `hardware_efficient_vqe` with depth chosen to match $\geq$ that many parameters.
- Recorded algorithm-reported iterations, wall time, final energy vs CASCI, and total cost-function evaluations wrapped counter for a fair trainability comparison.
- Accuracy vs number of parameters: `qubit_adapt_vqe` with a modular k -list: one run up to $\max(k)$ greedy operators; sliced energies at each k vs HF/CASCI, plus cumulative cost-eval and runtime snapshots.
- defaults use active-space Hamiltonians without CS so ADAPT can grow from HF CS optional where noted in docs.

2.6 Important Terms

L2 distance, also known as Euclidean distance, refers to the straight-line distance between two points; in our context, it refers to the distance between initial and final energy and parameter values. Cosine similarity measures the cosine of the angle between two vectors. In our context, cosine similarity helps compare how energy and parameter drift change between our noisy optimum and noiseless baseline.

3 Results

First, we verified that the reference energies matched the hartree-fock energies, and then the VQEs were able to at least equal to the hartree-fock energies or a lower value. This was successfully done. A few of the plots can be seen in Figure 3.

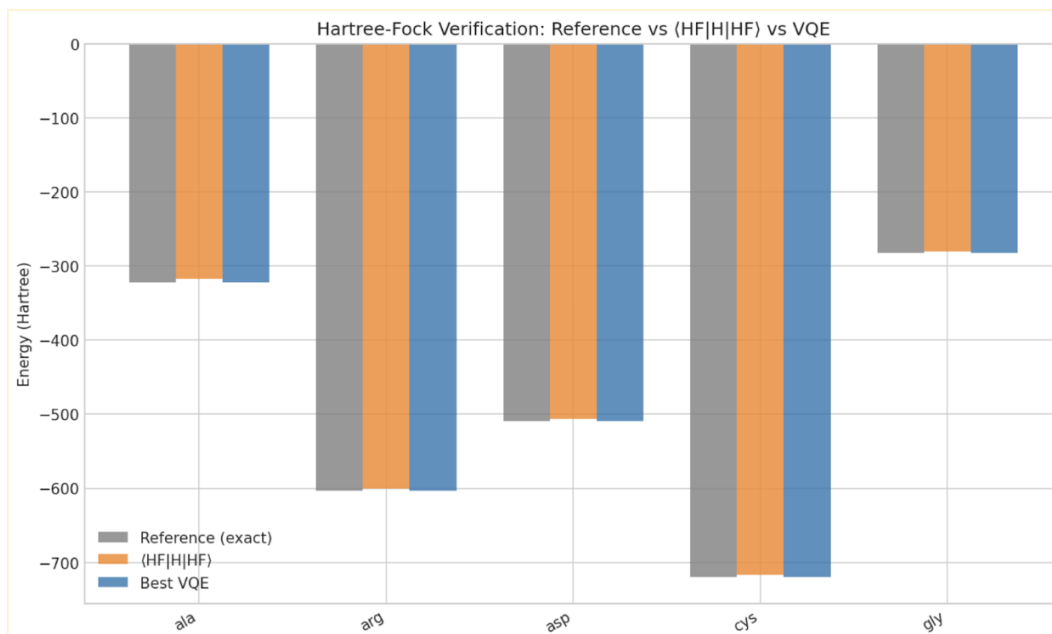


Figure 4: HF Verification VS other metrics on the 5 QMProt molecules.

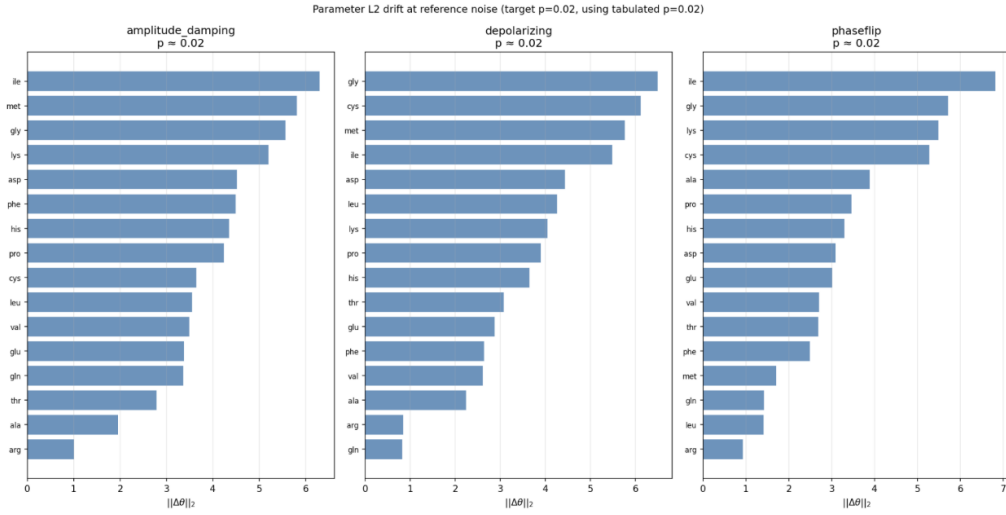


Figure 5: Parameter L2 Drift at Reference Noise. These, on their own, don’t tell you much about how significant the parameter drift is relative to baseline, but does tell you about relative drift between molecules. Refer to Figure 6 for further information. These were run on 36 parameters on 6-qubit, 2-layer RY hardware-efficient setup; an L2 norm of roughly 6 is compatible with “order-1 radian” average coordinate shifts. This was generated from experiment 1.

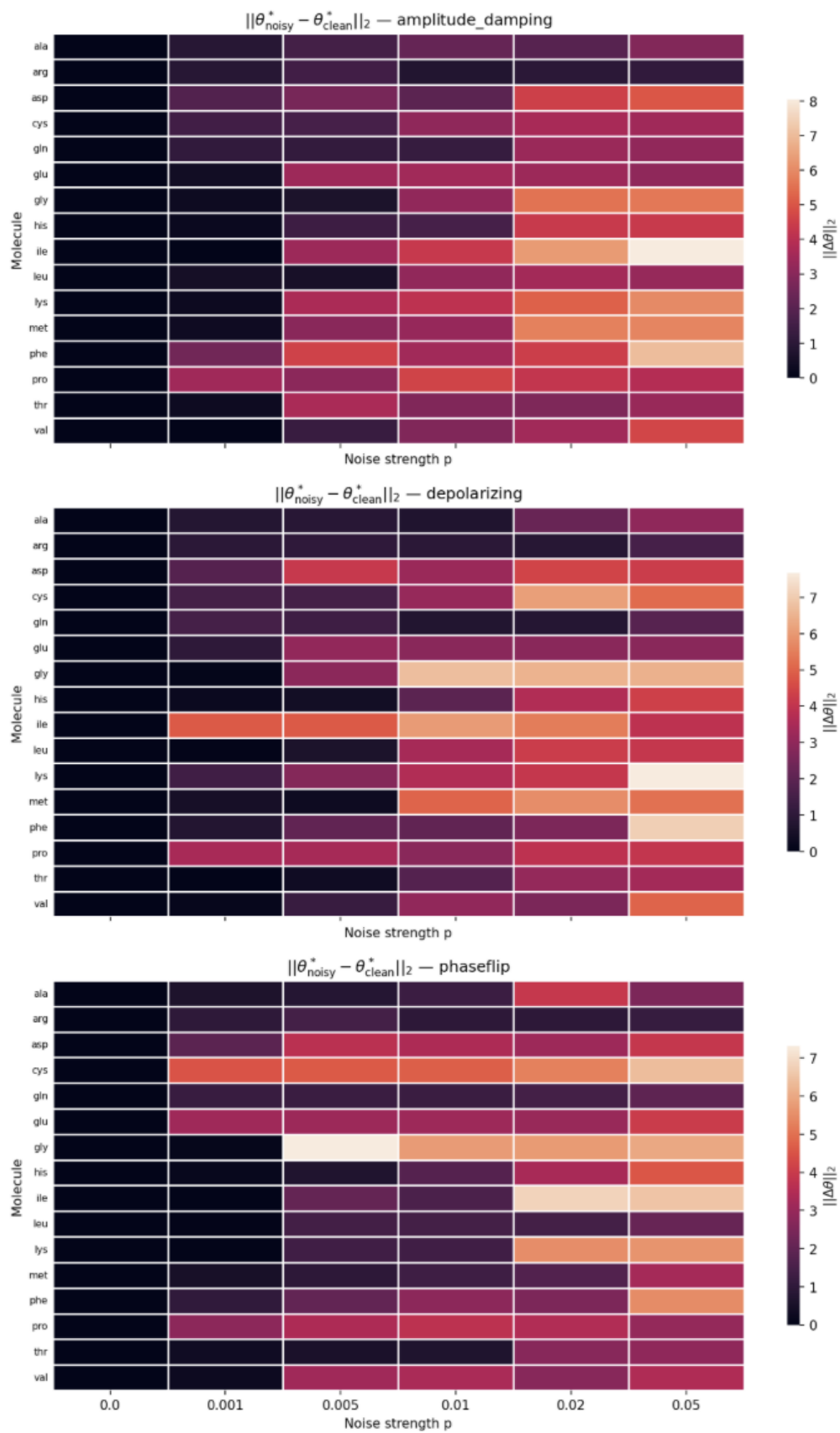


Figure 6: Heatmap of parameter drift measured using the L2 norm.

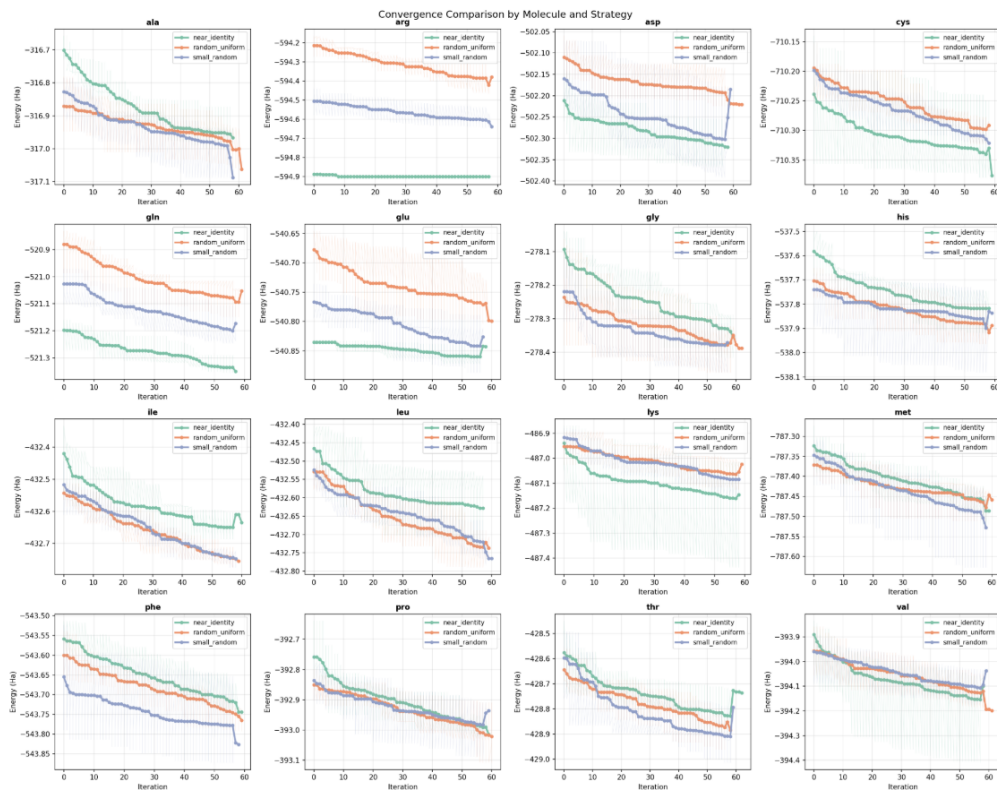


Figure 7: Investigating convergence for different strategies for amino acids from the dataset, using `hardware_efficient_vqe`.

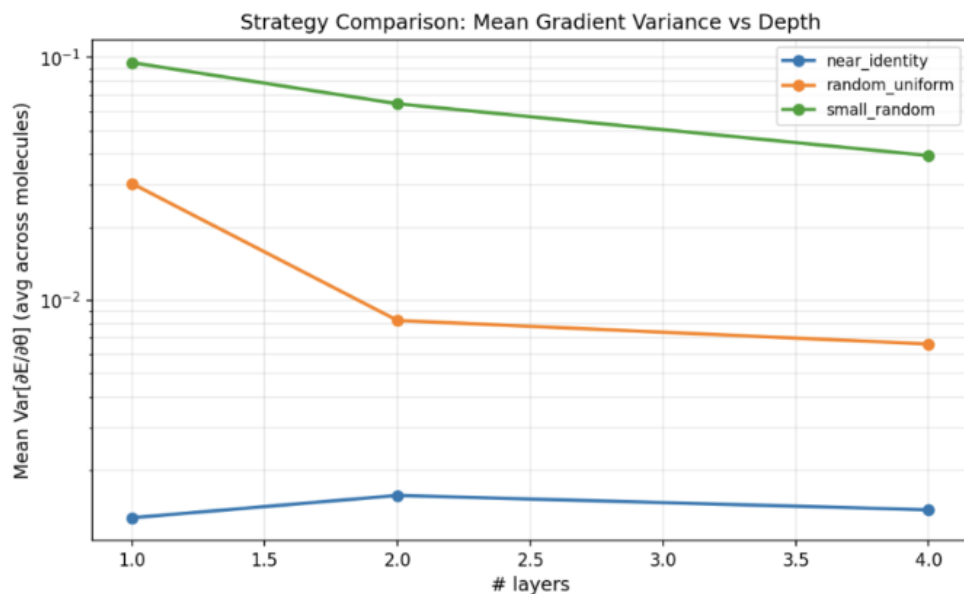


Figure 8: Comparing the mean variance across molecules, compared to the number of layers, for the three different strategies.

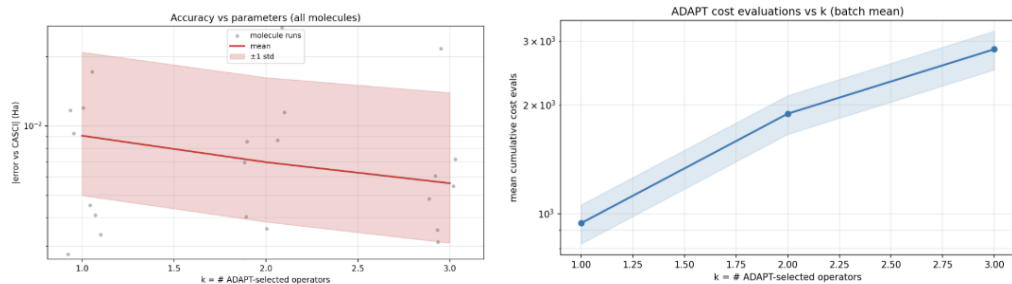


Figure 9: Investigating energy error vs $n_{\text{parameters}}$ for Adapt-VQE, compared to the CASCI. All of the amino acids in the QMProt dataset were run up to $k = 3$. Larger experiments can be run.

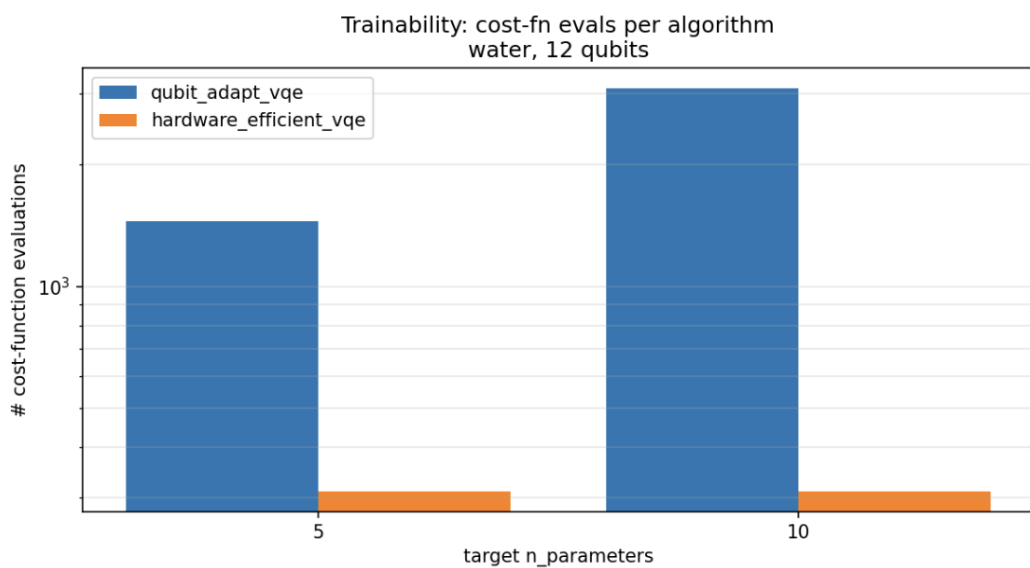


Figure 10: Investigating cost-fn evals per algorithm, just for water.

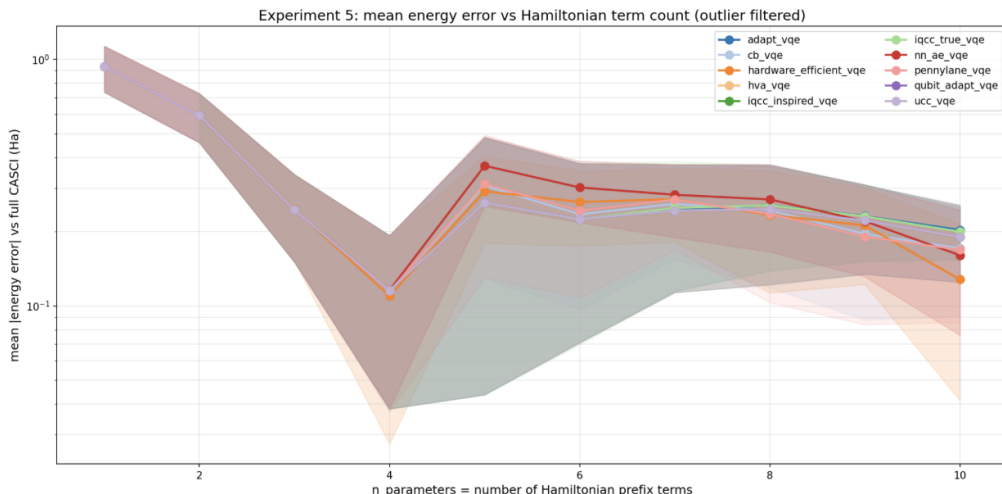


Figure 11: Investigating how much keeping the first n -parameters influences the mean energy error.

4 Discussion

We have created a repository for researchers to be able to run multiple VQE ansatz with multiple truncation methods in a fully automated pipeline. Researchers just need to install the requirements, and then they are able to run over 10 different ansatz, with automated active space truncation on their h5 files. This can be useful not only for biological molecules but also for other sorts of small to medium-sized molecules in order to better model near-term quantum hardware capabilities.

We’ve also investigated noise resilience for different molecules, barren plateaus and initialization, trainability (matched parameters), and accuracy as a function of the number of parameters, which will be informational for researchers deciding tradeoffs on running molecular simulations on real quantum hardware.

Furthermore, VQEs have a number of applications in fields such as quantum hardware development and lattice gauge theory (LGT). VQEs enable the simulation of quantum systems of such a large size that classical methods would demand too high of a computational cost, and quantum systems described by lattice gauge theory are no exception. Notably, working with LGT requires constrained optimization, in that one of the key goals is finding the lowest-energy eigenstate which satisfies the Gauss Law; algorithmic modifications are thus required, but the fact remains that VQEs facilitate working with LGT in a way classical methods do not. Therefore, our benchmarking system will facilitate the application of various VQEs to LGT problems, enabling researchers to better ascertain the effectiveness of different ansatz in LGT. Cheng notes that the development of variational ansatzes specifically tailored for LGT will be necessary to avoid the barren plateau problem in larger systems; nonetheless, our benchmarking system will be able to function as an important nexus for lattice gauge theory research.

It is possible that our results for the number of operations is overly dependent on the optimization algorithms that we’re using for the VQE, [10–12] so a future research direction or expansion of the repository can also compare those.

5 Code Availability

All reference code is available in the project repository.¹

¹https://github.com/sanskriti-ss/qmprot_gse

The repository includes the instructions for downloading the QMProt Dataset, and how to use the repository to run different VQE on your own hamiltonian datasets.

6 Appendix

(Preliminary exploration) Note that ANI energies are not variational (unlike the quantum chemistry methods), so they are not guaranteed to be above the true ground state. They might be too low.

Number of active orbitals selected using a frontier orbital window in STO-3G (RHF, frozen core), consistent with common truncated CAS approaches in quantum simulated benchmarks

A consistent rule here would be to select HOMO-1, HOMO, LUMO, LUMO+1, and additional orbitals if present.

<https://pubs.acs.org/doi/10.1021/acs.jctc.3c00123>

(Note: The orbital counts below are properties of our truncation strategy. Also, this is contingent on the blue steps described in the methodology being implemented.)

Amino Acid Active Orbitals Alanine (Ala) 4 Arginine (Arg) 6 Asparagine (Asn) 6 Aspartate (Asp) 6 Cysteine (Cys) 6 Glycine (Gly) 4 Proline (Pro) 6 Serine (Ser) 6 Valine (Val) 6

References

- [1] Laia Coronas Sala and Parfait Atchade-Adelomou. QMProt: A comprehensive dataset of quantum properties for proteins. *Electronics*, 14(14):2825, 2025. doi: 10.3390/electronics14142825. URL <https://www.mdpi.com/2079-9292/14/14/2825>.
- [2] John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018. doi: 10.22331/q-2018-08-06-79. URL <https://quantum-journal.org/papers/q-2018-08-06-79/>.
- [3] Juan Angel de Gracia Triviño, Mickael G. Delcey, and Göran Wendin. Complete active space methods for NISQ devices: The importance of canonical orbital optimization for accuracy and noise resilience. *Journal of Chemical Theory and Computation*, 19(10):2863–2872, 2023. doi: 10.1021/acs.jctc.3c00123. URL <https://doi.org/10.1021/acs.jctc.3c00123>.
- [4] Nikolaj Moll, Panagiotis Barkoutsos, Lev S. Bishop, Jerry M. Chow, Andrew Cross, Daniel J. Egger, Stefan Filipp, Andreas Fuhrer, Jay M. Gambetta, Marc Ganzhorn, Abhinav Kandala, Antonio Mezzacapo, Peter Müller, Walter Riess, Gian Salis, John Smolin, Ivano Tavernelli, and Kristan Temme. Quantum optimization using variational algorithms on near-term quantum devices. *Quantum Science and Technology*, 3(3):030503, 2018. doi: 10.1088/2058-9565/aab822. URL <https://doi.org/10.1088/2058-9565/aab822>.
- [5] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. Error mitigation for short-depth quantum circuits. *Physical Review Letters*, 119(18):180509, 2017. doi: 10.1103/PhysRevLett.119.180509. URL <https://arxiv.org/abs/1612.02058>.
- [6] Qing Guo and Ping-Xing Chen. Optimization of VQE-UCC algorithm based on spin state symmetry. *Frontiers in Physics*, 9:735321, 2021. doi: 10.3389/fphy.2021.735321. URL <https://doi.org/10.3389/fphy.2021.735321>.
- [7] Qiming Sun, Jun Yang, and Garnet Kin-Lic Chan. A general second order complete active space self-consistent-field solver for large-scale systems. *Chemical Physics Letters*, 683:291–299, 2017. doi: 10.1016/j.cplett.2017.03.004.
- [8] PySCF Developers. PySCF user guide: Multi-configuration self-consistent field (MCSCF). <https://pyscf.org/user/mcscf.html>, 2026. Accessed May 24, 2026.

- [9] William M. Kirby, Andrew Tranter, and Peter J. Love. Contextual subspace variational quantum eigensolver. *Quantum*, 5:456, 2021. doi: 10.22331/q-2021-05-14-456. URL <https://quantum-journal.org/papers/q-2021-05-14-456/>.
- [10] Dominic Alfonso, Yueh-Lin Lee, Hari P. Paudel, and Yuhua Duan. Chemical applications of variational quantum eigenvalue-based quantum algorithms: Perspective and survey. *Applied Physics Reviews*, 12(3):031304, 2025. doi: 10.1063/5.0245874. URL <https://doi.org/10.1063/5.0245874>.
- [11] Li-Xiang Cheng and Dan-Bo Zhang. Variational quantum simulation of ground states and thermal states for lattice gauge theory with multi-objective optimization. *Physics Letters A*, 546:130516, 2025. ISSN 0375-9601. doi: 10.1016/j.physleta.2025.130516. URL <https://doi.org/10.1016/j.physleta.2025.130516>.
- [12] Kim A. Nicoli, Luca J. Wagner, and Lena Funcke. Machine-learning-enhanced optimization of noise-resilient variational quantum eigensolvers, 2025. URL <https://arxiv.org/abs/2501.17689>.